

CHAPTER – 1A

TRAFFIC ACCIDENTS

Introduction

India has a well-knit and coordinated system of transport which plays an important role in development of economic activities by promoting fair distribution of produced goods and services and movement of people. The share of transport sector in Gross Domestic Product (GDP) of India is steadily growing. It is one of the key indicators in assessment of socio-economic development of the country. Since traffic accidents are indicator of bottlenecks and other hindrances in smooth flow of traffic, hence NCRB collects detailed data on traffic accidents including road accidents for inferring the trend and patterns of traffic accidents for the planners to devise appropriate preventive strategies.

The Bureau collects data on 'Traffic Accidents' comprising of (i) Road Accidents (ii) Railway Accidents and (iii) Railway Crossing Accidents, as these are the major contributors of accidental deaths.

Number of 'Traffic Accidents' in the country have increased from 4,22,659 in 2021 to 4,72,467 in 2022. (The rate of deaths in road accidents per thousand vehicles has decreased from 0.53 in 2021 to 0.52 in 2022). Maximum increase in number of traffic accidents cases in States from 2021 to 2022 was reported in Sikkim (from 122 to 199) followed by Manipur (from 366 to 510), Kerala (from 33,051 to 43,970), D&N Haveli and Daman & Diu (from

140 to 175) and Uttarakhand (from 1,430 to 1,749). [Table-1A.1].

These traffic accidents resulted in injuries to 4,25,727 persons and 1,94,347 deaths during 2022. State of Uttar Pradesh (28,615 deaths) followed by Tamil Nadu (19,717 deaths) and Maharashtra (18,893 deaths) have reported maximum fatalities in traffic accidents in the country; these 3 States accounted for 14.7%, 10.1% and 9.7% of total deaths in traffic accidents respectively and collectively accounted for 34.6% (67,225 out of 1,94,347) of total fatalities reported at all India level during 2022.

The percentage share of traffic accidental deaths in total deaths due to 'Other Causes' has increased from 44.2% in 2018 to 46.0% in 2022. A rising trend was seen in total number of deaths in 'Traffic Accidents' from 2018 to 2019 with sharp decline in 2020, which has again increased in 2021 but this number in 2021 is less than that in 2018 & 2019. In 2022, it raised to 1,94,347. Total number of Accidental deaths have increased by 11.8% (from 1,73,860 to 1,94,347) in 2022 over 2021. [LIST-1A.1].

These traffic accidents comprise of 4,46,768 road accidents, 23,139 railway accidents and 2,560 railway crossing accidents and caused 1,71,100, 20,792 and 2,455 deaths respectively during 2022.

LIST-1A.1

Number and Share of Deaths due to Traffic Accidents during 2018 – 2022

Sl. No.	Year	Number of Accidental Deaths					Percentage Share of 'Traffic Accidental Deaths' in Accidental Deaths due to 'Other Causes'
		Road Accidents	Railway Accidents	Railway Crossing Accidents	Total Traffic Accidents	Total Accidental Deaths due to 'Other Causes'	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1	2018	1,52,780	24,545	1,507	1,78,832	4,04,933	44.2%
2	2019	1,54,732	24,619	1,762	1,81,113	4,12,959	43.9%
3	2020	1,33,201	11,968	1,185	1,46,354	3,66,992	39.9%
4	2021	1,55,622	16,431	1,807	1,73,860	3,90,404	44.5%
5	2022	1,71,100	20,792	2,455	1,94,347	4,22,444	46.0%

- As per data provided by States/UTs.

FIGURE-1A.1
STATE/UT – WISE TRAFFIC ACCIDENT CASES DURING 2022



Traffic Accident Cases

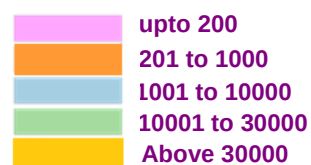
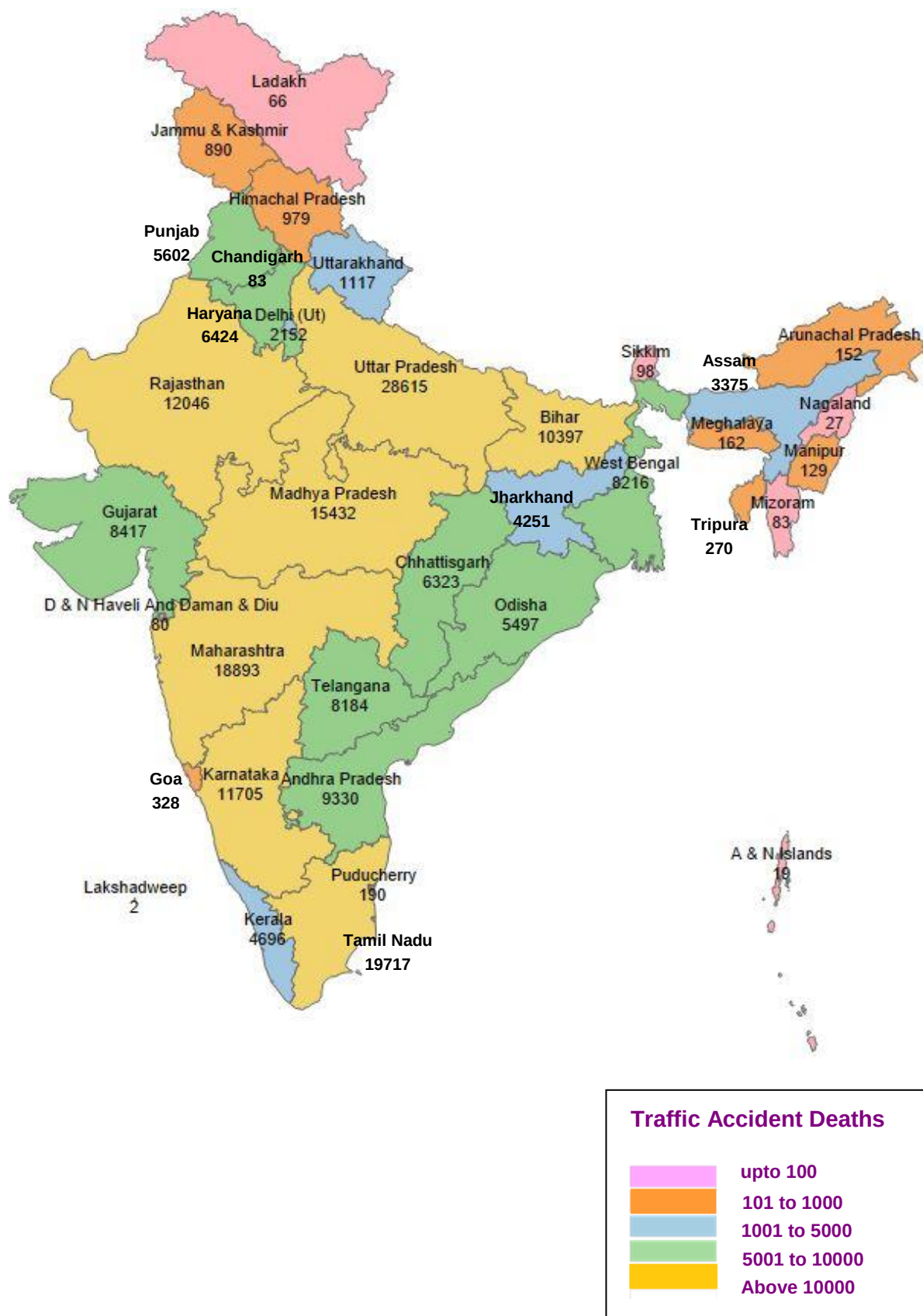


FIGURE-1A.2
STATE/UT – WISE TRAFFIC ACCIDENT DEATHS DURING 2022



- As per data provided by States/UTs.

Month - wise patterns of traffic accidents reveal that maximum number of 'Traffic Accidents' have occurred in the month of May which accounted for 9.2%.The month-wise break-up of 'Traffic Accidents' for each State/UT/City are presented in **Table-1A.5**.

Time of occurrence - wise analysis of traffic accidents reveal that maximum number of 'Traffic Accidents' have taken place during 18:00 hrs – 21:00 hrs and 15:00 hrs – 18:00 hrs, accounting for 20.0% (94,460) and 17.1% (80,855) of total traffic accidents (4,72,467) respectively during the year 2022. State/UT-wise 'Traffic Accidents' by time of occurrence is presented in **Table-1A.6**.

Road Accidents

The Bureau has made an effort to capture comprehensive information on road accidents using the revised proformae and published the first report for the year 2014 and this edition is ninth in the series.

A total of 4,46,768 road accident cases were reported during 2022. Road accident cases in the country have increased from 4,03,116 in 2021 to 4,46,768 [**Table-1A.1**]. The fatalities in road accidents have increased by 9.9% (from 1,55,622 in 2021 to 1,71,100 in

2022). The **LIST-1A.1** can be referred to see the patterns of 'Road Accidental Deaths'.

The number of vehicles, number of road accidents along with resultant fatalities and injuries therefrom, their percentage variations over previous year and the rate of accidental deaths per thousand vehicles during the last five years are presented in **LIST-1A.2**. It is observed that the rate of deaths per thousand vehicles in 2022 has decreased from 0.53 in 2021 to 0.52 in 2022.

4,46,768 road accidents caused 1,71,100 deaths and injuries to 4,23,158 persons during 2022. Generally road accidents have caused more injuries than deaths, but in Mizoram, Jharkhand, Punjab, Bihar and Uttar Pradesh, road accidents caused more deaths compared to persons injured. In Mizoram, 71 road accidents caused 83 deaths & injuries to 28 persons, in Jharkhand, 5,175 road accidents caused 3,898 deaths and injuries to 3,747 persons, in Punjab, 6,122 road accidents caused 4,688 deaths and injuries to 3,372 persons, in Bihar, 10,801 road accidents caused 8,898 deaths and injuries to 7,068 persons, and in Uttar Pradesh, 36,875 road accidents caused 24,109 deaths and injuries to 21,696 persons. [**Table-1A.2**].

LIST-1A.2
Growth in Number of Vehicles and Road Accidents in India (2018-2022)

Sl. No.	Year	Road Accidents (in thousand)	% Variation over Previous Year	Persons Injured (in thousand)	% Variation over Previous Year	Persons died (in Nos.)	% Variation Over Previous Year	No. of Vehicles (In Thousand)#	% Variation over previous Year	Rate of Deaths per thousand Vehicles (Col.7/Col.9)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
1	2018	445.5	0.00	446.5	-2.10	1,52,780	1.80	2,72,587	7.6	0.56
2	2019	437.4	-1.8	439.2	-1.6	1,54,732	1.3	2,95,772	8.5	0.52
3	2020	354.8	-18.9	335	-23.7	1,33,201	-13.9	2,95,772*	-	0.45
4	2021	403.1	13.6	371.9	11.0	1,55,622	16.8	2,95,772*	-	0.53
5	2022	446.8	10.8	423.2	13.8	1,71,100	9.9	3,26,299	-	0.52

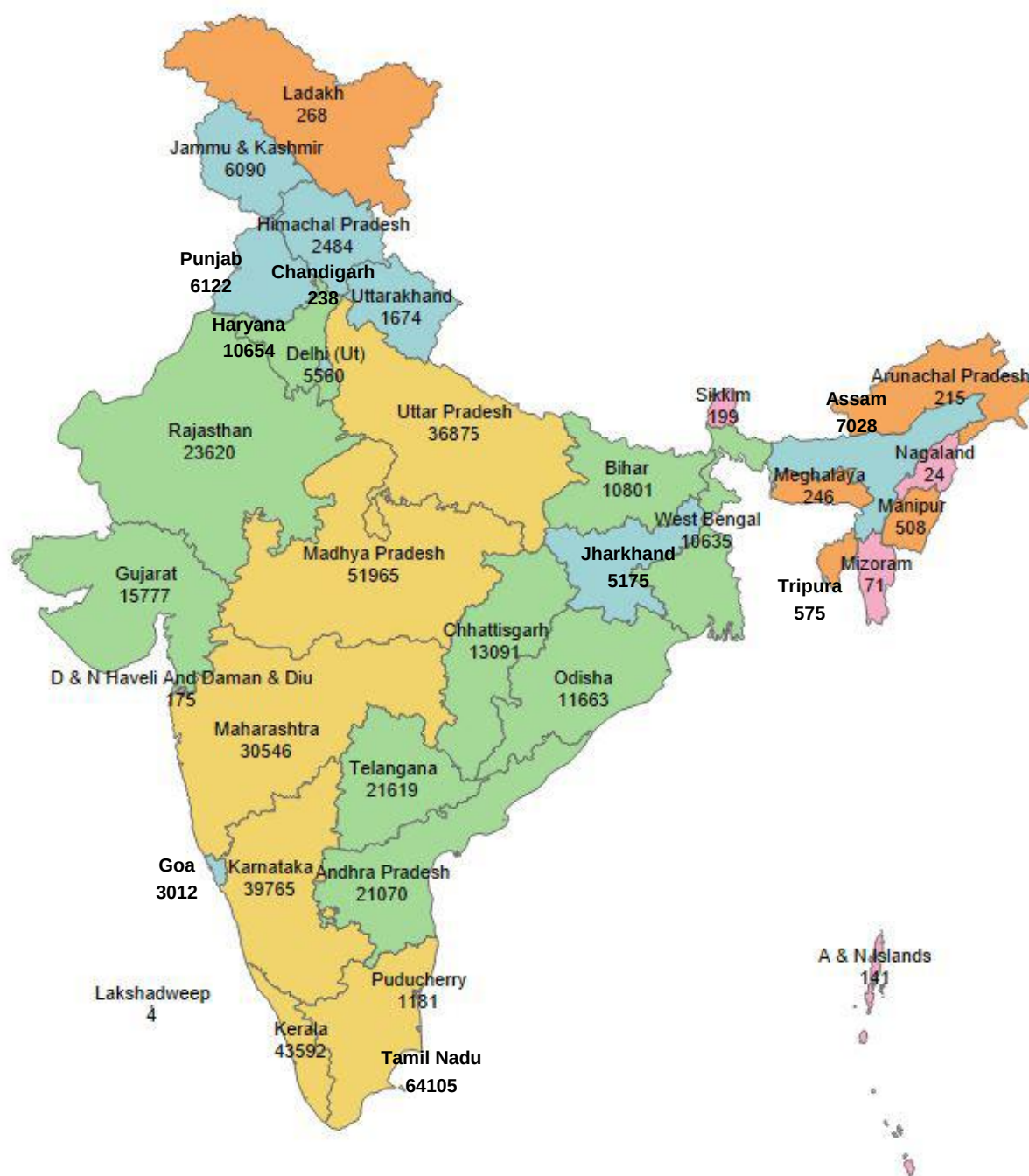
*# Source: Road Accidents in India - 2021, TRW, MoRT&H, as per latest published data.

*# figures of the previous year used due to non-availability of data.

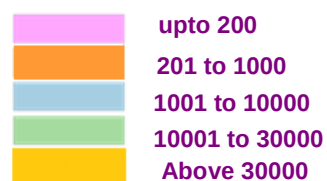
‘+’ for calculation, figures of previous year regarding number of vehicles has been used due to unavailability of data.

- As per data provided by States/UTs.

FIGURE-1A.3
STATE/UT – WISE ROAD ACCIDENT CASES DURING 2022



Road Accident Cases



- As per data provided by States/UTs.

During 2022, two wheelers have accounted for maximum fatal road accidents (77,876 deaths), contributing 45.5% of total road accidental deaths, followed by Pedestrian (24,742 deaths) (14.5%) and Cars (24,086 deaths) (14.1%) [Table-1A.3].

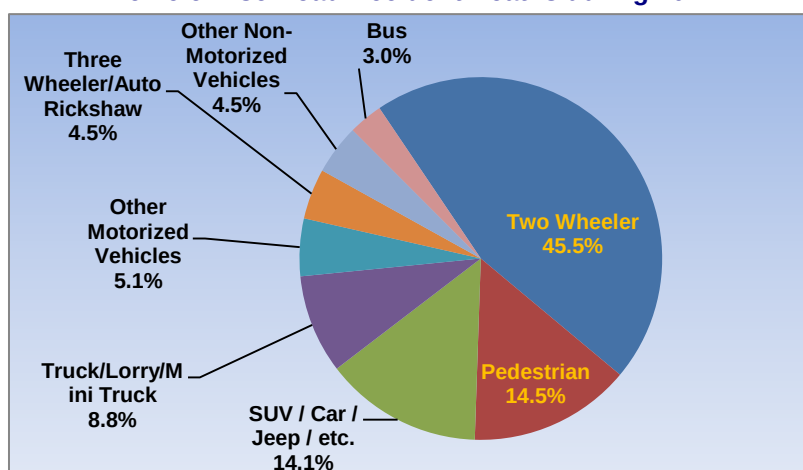
Majority of deaths due to two wheelers accidents were reported in Tamil Nadu (11,140 deaths) and Uttar Pradesh (8,305 deaths), accounting for 14.3% and 10.7% of total deaths due to two wheeled vehicles respectively. Large number of deaths due to SUV/car/jeep accidents (4,337 out of 24,086) were reported in Uttar Pradesh (18.0%) of total such accidents and large number of deaths due to trucks/lorries/mini truck accidents (4,109 out of 15,087) were reported in Uttar Pradesh, accounting for 27.2%. 33.5% (1,744 out of

5,211) and 7.3% (380 out of 5,211) of total fatal road accidents due to buses were reported in Uttar Pradesh and Madhya Pradesh respectively. Deaths of 17.9% (4,427 out of 24,742) of pedestrians in road accidents were reported in Tamil Nadu during 2022 [Table-1A.4].

The month-wise distribution of 'Road Accidents' shows that most of road accidents were reported in the month of May 41,298 cases), contributing 9.2% of total road accidents. Majority of accidents in this month (May) have been reported in Tamil Nadu, accounting for 13.4% of total accidents reported (5,546 out of 41,298 cases) in the month of May [Table-1A.5].

FIGURE-1A.4

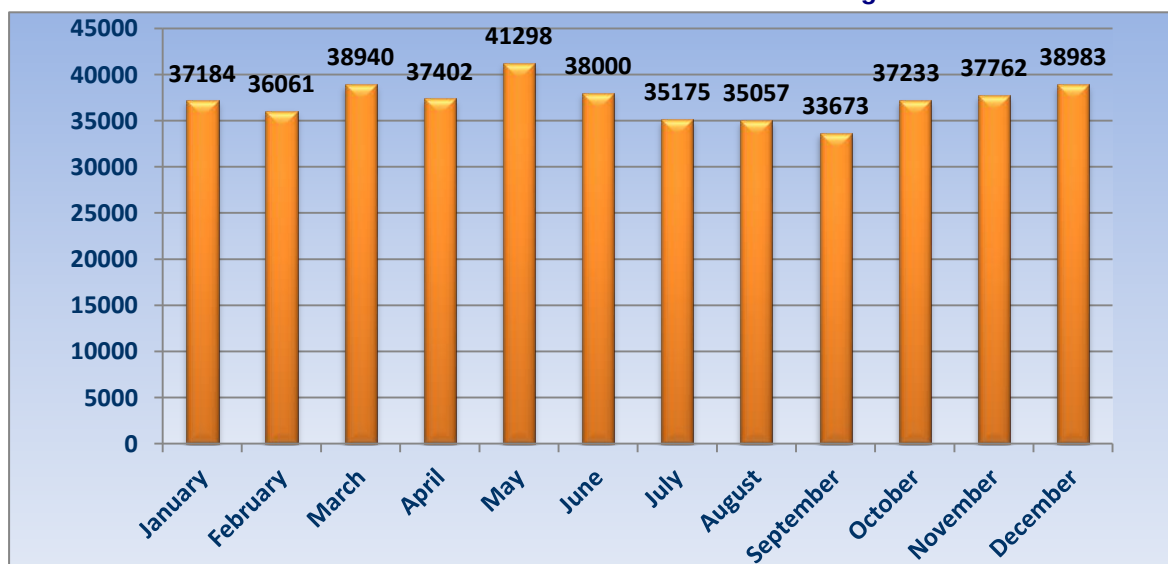
Vehicle wise Road Accident Deaths during 2022



Other Motorised Transport includes Tractor etc.

FIGURE-1A.5

Month-wise occurrence of Road Accidents during 2022



- As per data provided by States/UTs.

Most of road accidents (90,663 out of 4,46,768 cases) were reported during 18:00 hrs to 21:00 hrs (Night), accounting for 20.3% of total road accidents. During 18:00 hrs to 21:00 hrs (Night), majority of road accidents were reported in Tamil Nadu (16,285 cases), Madhya Pradesh (9,907 cases) and Kerala (9,089 cases). Time period '15:00 hrs to 18:00 (Day)' and '12:00 hrs to 15:00 hrs (Day)' accounted for 17.3% (77,514 cases) and 14.9% (66,756 cases) of total road accidents during 2022 [Table-1A.6].

Road-wise classification of accidents is presented in Table-1A.7. As per road classification, the National Highways which has a share of only 2.1% of total road length* (1.33 Lakh KMs out of 63.3 Lakh KMs) accounted for highest road accidents, contributing 30.5% of total road accidents. State Highways having the share of 2.8% (1.80 Lakh KMs) of total road length have reported 23.8% of road accidents in the country. However, a considerable number of road accidents were also reported on other roads. These accounted for 45.8% of total such accidents during 2022.

A total of 2,446 cases of road accidents were also reported on Expressways which caused injuries to 1,651 persons and deaths of 1,780 persons. The highest numbers of deaths in road accidents were reported on

the National Highways accounting for 34.9% (59,673 out of 1,71,100) followed by State Highways (24.5%) (42,003 deaths). A total of 69,424 (40.6%) persons died due to road accidents on the other roads during 2022.

State/UT-wise patterns revealed that maximum fatalities in road accidents on the National Highways took place in Uttar Pradesh (13.8%) (8,236 out of 59,673 deaths) followed by Tamil Nadu (10.0%) (5,978 deaths), Maharashtra (8.1%) (4,833 deaths), Karnataka (7.0%) (4,201 deaths) and Rajasthan (6.7%) (3,997 deaths) during 2022.

Maximum number of accidents on State Highways in the country occurred in Tamil Nadu (22,633 cases). Maximum fatalities in road accidents on State Highways were reported in Tamil Nadu (6,364 out of 42,003 deaths) which accounted for 15.2% of total deaths due to road accidents on State Highways, followed by Uttar Pradesh (6,070) (14.5%) during 2022. Maximum fatalities on the Expressways were reported in Uttar Pradesh contributing 73.6% (1,310 out 1,780) followed by Haryana (6.0%), West Bengal (5.7%), Maharashtra (4.9%) and Gujarat (3.6%) during 2022. [Table-1A.7]

LIST-1A.3 Road Accidental Deaths in every 100 km-2022 (Road Category wise)

Type of Road	2022				
	Length of Road (in KM*)	Road Accident Cases	Total Number of Deaths	Cases per 100 KM	Deaths per 100 KM
National Highways	1,32,499 [#]	1,36,122	59,673	103	45
State Highways	1,79,535 [#]	1,06,155	42,003	59	23
Other Roads	60,19,723 [#]	2,04,491	69,424	3	1
Total	63,31,757 [#]	4,46,768	1,71,100	7	3

- As per data provided by States/UTs.

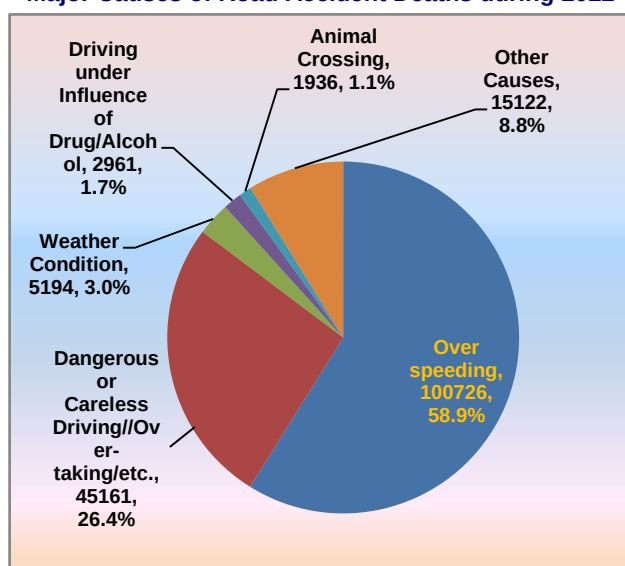
* Source: Road Accidents in India – 2021, Ministry of Road Transport & Highways.

figures of the previous year used due to non-availability of data.

Cause-wise distribution of road accidents (which also include unmanned railway crossing accidents) is presented in **Table-1A.8**. Cause-wise analysis of road accidents revealed that most of road accidents were due to over-speeding accounting for 62.6% of total accidents (2,79,674 out of 4,46,781 cases) which caused 1,00,726 deaths and injuries to 2,71,661 persons. Dangerous/careless driving or overtaking caused 1,10,440 accidents (24.7% of total accidents) which resulted in 45,161 deaths and injuries to 1,00,901 persons during 2022. 2.2% (9,973 out of 4,46,781 cases) of such accidents were due to poor weather condition. Driving under influence of drug/alcohol contributed 1.6% of total accidents which resulted in injuries to 6,320 persons & 2,961 deaths in the country.

FIGURE–1A.6

Major Causes of Road Accident Deaths during 2022



- As per data provided by States/UTs.

Cause - wise analysis of fatal road accidents revealed that 58.9% (1,00,726 out of 1,71,100 deaths) and 26.4% (45,161 out of 1,71,100 deaths) of fatalities in road accidents were due to over-speeding and dangerous/careless driving or overtaking respectively. Poor weather conditions, Driving under Influence of Drug/Alcohol and Animal Crossing caused 3.0% (5,194 deaths), 1.7% (2,961 deaths) and 1.1% (1,936 deaths) of total deaths due to road accidents respectively during 2022.

Large number of deaths in road accidents due to over-speeding was reported in Tamil Nadu (14.0%) (14,086 out of 1,00,726 deaths) followed by Maharashtra (10.0%) (10,162 out of 1,00,726 deaths). Dangerous/careless driving or overtaking caused maximum fatalities in Uttar Pradesh (13,171 out of 45,161) which accounted for 29.2% of total deaths followed by 8.9% (4,015) deaths in Maharashtra. Maximum fatalities due to driving under influence of drug/alcohol were reported in Uttar Pradesh (31.4%) followed by Odisha (11.2%), Madhya Pradesh (9.3%), Maharashtra (7.2%) and Telangana (6.4%) during 2022 [**Table-1A.9**].

A total of 13 accidental deaths were reported at un-manned railways crossings. All the 13 accidental deaths were reported in Uttar Pradesh during 2022 [**Table-1A.9**].

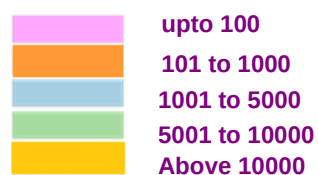
Place of occurrence - wise patterns of road accidents reveal that 59.7% of total accidents have occurred in rural areas (2,66,707) and 40.3% in urban areas (1,80,061) out of 4,46,768 total cases during 2022. Both in rural as well as urban area most of the accidents were reported at places near residential area. 30.2% (80,536 out of 2,66,707 cases) accidents in rural area and 28.5% (51,257 out of 1,80,061 cases) in urban area have taken place near residential area. 7.0% of road accidents in urban area took place at pedestrian crossing (12,515 out 1,80,061 cases). Besides, 6.9% (30,971 out of 4,46,768 cases) of road accidents in the country have taken place near schools, college or other educational institutions [**Table 1A.10**].

Uttar Pradesh followed by Maharashtra have reported 27.0% and 10.2% of total deaths due to road accidents near schools/college/other educational institutes in urban area respectively. Uttar Pradesh reported highest fatalities due to road accidents at places near to residential area (urban area) accounting for 17.0% (2,866 out of 16,850) of total such deaths during 2022 [**Table-1A.11**].

FIGURE-1A.7
STATE/UT – WISE ROAD ACCIDENT DEATHS DURING 2022



Road Accident Deaths



- As per data provided by States/UTs.

Railway Accidents

A total of 23,139 cases of 'Railway Accidents' were reported, showing an increase of 28.6% during the year 2022 over 2021 (17,993). 23,139 railways accidents caused injuries to 2,457 persons and 20,792 deaths during 2022 [Table-1A.1 & 1A.2].

Maximum railway accidents were reported in Maharashtra accounting for 20.9% (4,827 out of 23,139 cases) followed by Uttar Pradesh (13.8%) (3,190 cases). These two States have also reported highest fatalities in railways accidents, accounting for 15.2% (3,166 out of 20,792 deaths) in Uttar Pradesh and 15.1% (3,145 deaths) in Maharashtra of total deaths in railways accidents respectively. 1,686 out of 2,457 persons injured in railways accidents were reported in Maharashtra alone during 2022 [Table 1A.2].

The month-wise distribution of 'Railway Accidents' shows that most of railway accidents were reported in the month of October (2,154 cases), contributing 9.3% of total railway accidents. Maharashtra (497 out of 2,154 cases) has reported maximum railways accidents in the month of October, accounting for 23.1% of total such accidents [Table-1A.5].

Most of railway accidents (3,819 out of 23,139) have taken place during 06:00 hrs to 09:00 hrs (Day), accounting for 16.5% of total railway accidents. 15.3% (3,534 cases) railways accidents were reported during '09:00 hrs to 12:00 hrs (Day)'. Maharashtra has reported maximum accidents during 18:00 hrs to 21:00 hrs (Night) and 09:00 hrs to 12:00 hrs (Day), accounting for 16.7% (805 cases) and 15.6% (755 cases) respectively [Table-1A.6].

State/UT - wise classification of railways accidents is presented in Table-1A.12 and Figure-1A.9. The analysis of classification of railway accidents revealed that incidents of 'Fall from Trains or Collision with People at Track' constituted majority of railway accidents (73.7%) (17,053 out of 23,139). State of Maharashtra has reported the majority of such cases, accounting for 27.4% (4,677 out of 17,053 cases) of total cases of fall from train or collision of trains with people at track. A total of

15,096 persons died due to either fall from trains or collision of trains with people at tracks, accounting for 72.6% of total deaths in railway accidents (20,792 deaths).

State/UT - wise causes of railways accidents is presented in Table-1A.13. The causes include Fault of Driver, Sabotage, Signalmen's Error, Mechanical Failure and Other causes. Majority of States/UTs have furnished railways accidents under unclassified category 'Other Causes', a total of 22,160 out of 23,139 cases of railways accidents were furnished under 'Other Cause' (fall of persons from trains, persons coming under trains etc. come under this category). During 2022, a total of 41 cases of railways accidents occurred due to fault of driver. Mechanical defects (like poor design, track faults, bridge/tunnel collapse, etc.) caused 688, 15, 71, 13, and 116 railways accidents in Assam, Bihar, Jharkhand, Kerala and Uttar Pradesh respectively. In Jharkhand, a total of 21 persons died in railways accidents due to fault of driver. Mechanical defects (like poor design, track faults, bridge/tunnel collapse, etc.) led to loss of 907 lives in railways accidents during 2022.

Maximum railway crossing accidents were reported in Uttar Pradesh accounting for 52.3% (1,340 out of 2,560 cases) followed by West Bengal (21.3%) (546 cases) and Madhya Pradesh (9.3%) (239 cases). States of Uttar Pradesh, West Bengal and Madhya Pradesh have reported highest fatalities in railway crossing accidents, accounting for 55.0% (1,340 out of 2,455 deaths), 20.1% (493 deaths) and 9.7% (239 deaths) respectively during 2022 [Table-1A.2].

Traffic Accidents in Cities

A total of 68,236 traffic accidents were reported in 53 cities during 2022. 68,236 traffic accidents caused injuries to 57,246 persons and 17,680 deaths. The maximum fatalities in traffic accidents was reported in Delhi City (2,103 deaths) followed by Jaipur (918 deaths) and Kanpur (795 deaths) [Table-1A.2].

Road Accidents (66,402 cases) accounted for 97.3% of total traffic accidents in 53 mega cities during 2022. Delhi (City) accounted for 8.1% (5,387 out of 66,402 cases) of total road

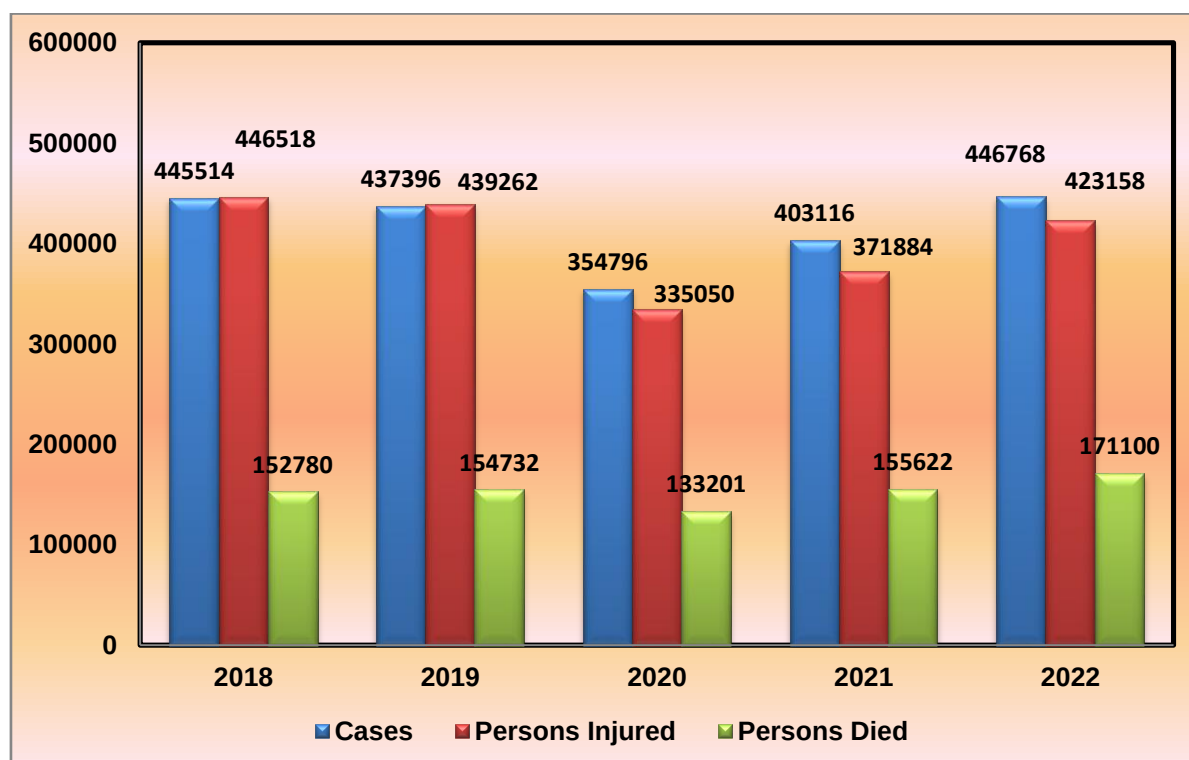
accidents reported in 53 mega cities followed by Bengaluru (5.8%) (3,822 cases) and Chennai (5.2%) (3,452 cases). The large number of fatal road accidents were reported in Delhi City (1,412 deaths) followed by Bengaluru (772 deaths), accounting for 8.8% and 4.8% of total deaths due to road accidents in 53 mega cities respectively during 2022.

Cause-wise analysis of road accidents (including unmanned railway crossing) revealed that most of road accident deaths (including unmanned railway crossing) in 53 mega cities were due to over-speeding which accounted for 51.5% (8,223 out of 15,968 deaths) of total deaths due to road accidents during 2022. Dangerous or Careless driving/Over-taking etc. also caused 31.3% of total deaths due to road accidents (4,990 out of 15,968 deaths). Driving under influence of drug/alcohol had caused 2.7% (427 out of 15,968 deaths) of fatalities in road accidents. Among 53 mega cities, most of fatalities due to driving under influence of drug/alcohol were reported in Prayagraj (104 out of 427 deaths) [Table-1A.9].

Place of occurrence – wise deaths in road accidents reveals that most of fatalities due to road accidents have taken place near residential area, contributing 29.1% (4,651 out of 15,967 deaths) of deaths in road accidents in 53 mega cities, followed by 9.7% (1,543 out of 15,967 deaths) near factory/industrial area and 8.7% near schools/ college/ other educational institutions (1,395 out of 15,967) deaths. Out of 53 mega cities, Ahmedabad (194 deaths) followed by Hyderabad (103 deaths) have reported maximum cases of road accidents at pedestrian crossing. [Table-1A.11]. As per road-wise classification of road accidents, 21.0% of total road accidents in 53 mega cities were reported at the National Highways. 28.6% of fatalities in road accidents were reported at the National Highways during 2022 [Table-1A.7].

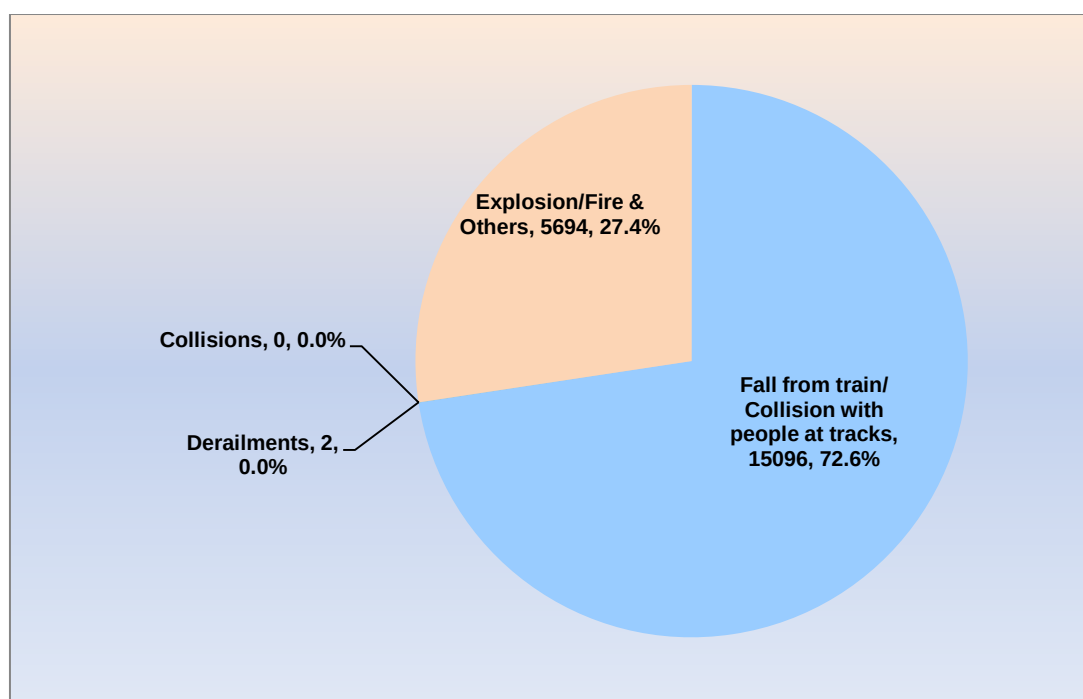
A total of 1,563 railway accidents were reported in 53 mega cities where in Delhi City (796) has reported maximum incidents by contributing 50.9% of total railway accidents during 2022.

FIGURE-1A.8
Trend of Road Accident Cases, Persons Injured and Persons Died during 2018–2022



- As per data provided by States/UTs.

FIGURE-1A.9
Classification of Railway Accident Deaths during 2022



- As per data provided by States/UTs.
